

Mark Farinas

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Technical Skills

Languages: Python, Kotlin, Dart, SQL, JavaScript, HTML/CSS, GDScript, C, VBA

Frameworks & Libraries: Jetpack Compose, Flutter, Ktor, JetBrains Exposed, Next.js, React, FastAPI, Retrofit

Databases & Infrastructure: PostgreSQL, PostGIS, SQLite, Supabase, Microsoft Access, NocoDB, Docker

Cloud & DevOps: Cloudflare (Workers, DNS, Email Routing), Vercel, Railway.io, Firebase, GitHub Actions, DNS/TLS administration

AI/ML: Ollama, Groq, RAG pipelines, vector embeddings, Whisper AI, LLM evaluation, agent-trajectory analysis, inter-rater agreement (Cohen's κ , Krippendorff's α)

Quality & Tools: pytest, mypy --strict, CI test matrices, Git, Figma

Experience

Cohere - Data Annotation Specialist, Software Engineering

Aug 2026 – Present

Independent Contractor

- Evaluate model-generated code and agent trajectories against rubrics for correctness, security, and edge-case handling; written justifications feed LLM training signal
- Debug AI-generated code across Python, JavaScript, Java, Go, and SQL; assess tool selection and error recovery in agent traces

Alberta Federation of Rural Water Coops - Junior Database Developer

Sep 2026 – Present

Edmonton, AB

- Retained on a fixed Sep–Dec 2026 term after the summer internship, promoted from Database Administrator Intern
- Own the conversion of the legacy Access database into a web-based system through cutover, user training, and handover
- Administrate the Water Federation website; automate form development and recurring administrative workflows

Alberta Federation of Rural Water Coops - Database Administrator Intern

May 2026 – Aug 2026

Edmonton, AB

- Built a re-runnable Python ETL pipeline (pyodbc to psycopg) migrating a legacy Access database to containerized PostgreSQL + NocoDB, validated by a 125-test fidelity suite
- Built TimeEntrySystem, a role-based Access/VBA timesheet app with a three-stage approval workflow and three-layer audit model for anti-fraud assurance
- Built InsuranceRenewalSystem, an Access/VBA renewal workflow matching five coverages by organization ID and generating 85 per-organization verification documents
- Coordinated migration rollout with stakeholders across rural Alberta water systems on data structure and access requirements

Alberta Precision Labs - COVID-19 Surveillance Research

Dec 2022 – Dec 2023

Edmonton, AB

- Conducted COVID-19 surveillance research using qPCR and dPCR across Albertan communities; ran daily RNA extraction and RT-PCR workflows

Popeye's Supplements - Customer Service

Jul 2022 – Present

Edmonton, AB

- Advise clients on supplements for athletic performance and wellness goals; run in-store demos and community education initiatives

Awards & Publications

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- 1st Place - CMPT 395 Software Engineering Competition** (IEEE-sponsored) *Apr 2026*
Won both the Semi-Finals and the Grand-Finals with Scheduler
- IEEE SmartEdu 2026** - co-authored full research paper *Accepted Jul 2026*
“Unjamming Deferrals: Supporting Students Through JAMN Bot, a Mobile RAG-Grounded Policy Assistant”

Education

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- MacEwan University** - BSc in Computer Science *Jan 2024 – Jun 2028*
Edmonton, AB
- University of Alberta** - BSc in Immunology and Infection *Sep 2016 – Apr 2021*
Edmonton, AB

Projects

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- Scheduler** - Backend & Android Developer, 5-person Agile/Scrum team *Jan 2026 – Apr 2026*
Kotlin, Ktor, JetBrains Exposed, SQLite, Jetpack Compose, Docker, Ollama
- Architected the Ktor + JetBrains Exposed + SQLite backend: JWT role authentication, room-booking engine with conflict detection and proctor load-balancing
 - Built role-specific student, faculty, and admin screens in Kotlin + Jetpack Compose over Retrofit, Firebase Auth, FCM, and DataStore
 - Designed automated FCM push triggers and a scheduled no-show poller in the backend service layer
 - Built an Ollama RAG pipeline for the AI policy assistant: nomic-embed-text embeddings, cosine-similarity retrieval over a SQLite vector store, role-aware prompt builder
 - Designed the evaluation behind the resulting IEEE SmartEdu paper: benchmark dataset, temperature ablations, human-vs-LLM-Judge inter-rater agreement
 - Containerized with Docker and deployed on Railway.io with environment-driven secrets management
- iMapped** - Web Core & Release Engineering, 5-person venture team *May 2026 – Present*
Flutter, Dart, Supabase, PostGIS, Groq, Vercel, GitHub Actions
- Built the web core of a trip-itinerary planner live at app.imapped.ca  - one Flutter codebase shipping to web and native Android, 172 tests green
 - Implemented the client-side route optimizer in Dart: haversine distance, nearest-neighbour construction, 2-opt improvement
 - Integrated Supabase (Postgres + PostGIS, PKCE auth, Edge Functions) behind repository interfaces, letting seed-backed fakes swap for live implementations without touching UI code
 - Deployed the recommendation re-ranker on Groq (Llama 70B) as a Supabase Edge Function, caching results to control latency and API cost
 - Led UI/UX and two rounds of user research, driving a pivot from flight booking to itinerary planning
 - Repositioned the product from tourist-only to everyday use with local loyalty mechanics
 - Built as a 5-person venture team in MacEwan’s technology-entrepreneurship course (COSL 301), owning the web core, landing page, and release engineering
 - Presented live twice - the course final and a stakeholder demo - covering market positioning, business model, and the deployed product
 - Carried the venture past the course: own the new GitHub organization, repository setup, and continuity of the live site
- Retail Storefront Rescue & Migration** - Solo client engagement *Jul 2026*
Cloudflare Workers, DNS/TLS, GoDaddy
- Rebuilt and relaunched a local retailer’s storefront overnight after a multi-day outage, reconnecting customers to its products at \$0/month recurring cost

- Executed the nameserver cutover with zero downtime for the client's online store: enumerated the legacy cPanel zone, verified DNSSEC state, preserved the e-commerce subdomain
- Rebuilt from scratch after finding the only surviving artifact was an unusable dev-mode SPA capture; delivered a dependency-free static site with no build step
- Self-directed the migration across unfamiliar platforms (GoDaddy, Cloudflare Workers, DNS, Email Routing), verifying provider terms and free-tier commercial-use limits first
- Configured TLS, Cloudflare Email Routing, and SPF/DKIM/DMARC (**p=reject**); added a fails-closed 18+ age gate and the missing privacy and terms pages
- Structured the handover around client ownership: client-held domain and admin access, a portable static-file deliverable, and a written runbook

VocabMaxxing - Android, Backend & UX Lead

Feb 2026 – Jul 2026

Kotlin, Jetpack Compose, Ktor, Groq, Firebase, Figma

- Implemented the Android frontend (login, dashboard, daily practice with microphone capture) in Kotlin + Jetpack Compose
- Built the Ktor + JetBrains Exposed + SQLite backend with Firebase Admin token verification and REST endpoints for auth and attempts; containerized on Railway.io
- Designed a Groq-powered (Llama 3.3 70B) semantic scoring service with a multi-dimensional rubric and hardened prompt-injection defenses
- Built a custom scoring engine combining morphological stemming and vocabulary-diversity heuristics, paired with the AI scorer in a hybrid pipeline
- Led ideation and UX design; produced Figma wireframes covering layout, navigation flow, and gamification mechanics (streaks, XP, coin shop)

Phone-Menu-Transcriber - Solo

Jun 2026

Python, Whisper AI, Ollama, FastAPI, Pydantic

- Paired Whisper speech-to-text with a locally-hosted Qwen 3 8B (Ollama) extractor behind a pluggable interface
- Produced Pydantic schema-constrained JSON, shipped as CLI and FastAPI service under `mypy --strict` with a four-version CI matrix

Maintenance Calorie Finder - Solo

Jul 2025

Python, SciPy, pytest

- Built a Python CLI that reverse-engineers maintenance calories from tracked intake and weight data
- Models metabolic adaptation over a goal window and solves for target intake with SciPy bounded optimization, validated by a 24-test pytest suite

Personal Portfolio - Solo

Dec 2025 – Present

Next.js, React

- Built a personal portfolio site in Next.js - design system inspired by American Psycho's business card aesthetic, client-side routing, responsive navigation

TiMED - Hack4Health 2025, co-designer

2025

Hardware + software concept, Figma

- Co-designed a hardware-software medication management system; wireframed the companion app (medication schedules, adherence reports, QR device pairing) and presented to judges